



Figure 3: Preferences of New Network

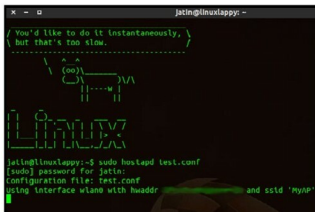


Figure 4: Hostapd running

connection between AP and laptop will #break. For CCMP, during rekeying, there is a break of approximately 10 seconds which may #be the time for random key generation and reconnection. #If you face any error you can skip the last option, but will work in most cases flawlessly
wpa_ptk_rekey=600
For more options refer to manual pages or Hostapd Website
<http://w1.fi/hostapd/>

Testing

We are now almost done but we need to create some arrangements. Using the command line would take a long time. The easy way would be to create a normal Ad-Hoc Wi-Fi hotspot (SSID not mandatory) to be the same as for Hostapd. Creating an Ad-Hoc hotspot is as easy as going to *Network Connections*, and then creating a *New Wireless Connection*.

Choose the SSID and key of your choice, because it doesn't matter, as this will be overridden by the WPA security of Hostapd.

In my case, the key is named *Hotspot*. After getting the notification that a successful connection has been established, fire up the terminal and type the following command:

```
sudo hostapd test.conf
```

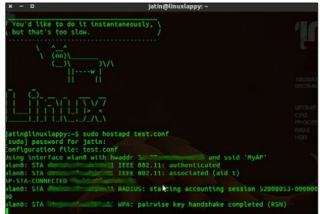


Figure 5: Connection details of your device shown by Hostapd

This command will invoke Hostapd with options specified by the configuration file named *test.conf* (which we created earlier). It will make a hotspot on the specified interface, *wlan0*, at the MAC address of your Wi-Fi card, as shown in Figure 4.

Now, try to connect your phone by specifying the necessary arguments. As soon as your phone tries to connect, you will see your device's MAC address along with its status, which would be something like what's shown in Figure 5.

Note: 1. Users have to first create an Ad-Hoc network, before issuing the Hostapd command.

Run the Hostapd command with *sudo* or with root privileges.

2. It would also help you to track who is connected to your network. MAC address filtering can be set up by just adding the following line:

```
#macaddr_acl sets options for mac address filtering. 0 means
"accept unless in deny list"
macaddr_acl=0
```



References

- [1] <http://ack100.erley.org/git.html>—for an explanation of how Hostapd manages Wi-Fi and an understanding of each component in the level.
- [2] <http://nims11.wordpress.com/2012/04/27/hostapd-the-linux-way-to-create-virtual-wifi-access-point/>—For an understanding of the rest of the process
- [3] <http://w1.fi/hostapd/>
- [4] <https://bugs.archlinux.org/task/27406> —for an explanation of the errors associated with wpa_ptk_rekey=600

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